

ScotJob Pulse AI — Technical Stack Document

Architecture Philosophy

Goals:

- Fast development
 - Low cost (£0–10/month initially)
 - Scalable later
 - Real-time notifications
 - Easy maintenance
 - Expandable to multiple job sources
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Frontend

MVP:

No frontend required initially.

Future dashboard:

Layer	Technology	Purpose
Framework	Next.js	Dashboard
UI	Tailwind CSS	Styling
Component s	shadcn/ui	UI components
Charts	Chart.js	Analytics

Backend

Layer	Technology	Purpose
Runtime	Node.js	Server runtime
Framework	Express.js	API/server
Language	TypeScript	Type safety

Reason:

- Fast development
- Huge package ecosystem
- Works well with scraping libraries

Scraping Layer

Tool	Purpose
Playwright	Dynamic pages
Axios	HTTP requests
Cheerio	HTML parsing

Strategy:

Fast path

Axios



Cheerio

Use for static pages

Dynamic path

Playwright

Use when JavaScript rendering is required

Scheduling Layer

Tool	Purpose
node-cron	Job scheduling

Example:

```
cron.schedule("* / 10 * * * *", checkJobs)
```

Runs every 10 seconds.

Notification Layer

Tool	Purpose
Telegram Bot API	Send alerts

Bot creation:

BotFather

↓

BOT_TOKEN

↓

Environment Variables

Database Layer

Development:

Tool	Purpose
SQLite	Local database

Production:

Tool	Purpose
PostgreSQL	Main DB

Tables:

Users
Jobs
JobLogs
Filters

ORM Layer

Tool	Purpose
Prisma	Database ORM

Benefits:

- Auto migrations
 - Type safety
 - Easier queries
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Caching Layer

Tool	Purpose
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[Redis](#) Temporary memory

Use:

- Store recent IDs
 - Prevent duplicates
 - Request throttling
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Monitoring + Logging

Tool	Purpose
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Winston	Logging
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Sentry	Error tracking
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Logs:

[10:00:03]

Amazon fetch started

[10:00:04]

3 jobs detected

[10:00:04]

Scotland filter passed

[10:00:05]

Telegram notification sent

Deployment Stack

Development:

Local machine

Production:

Tool	Purpose
Docker	Containerization
Railway	Hosting
Render	Alternative hosting

Environment Variables

BOT_TOKEN=xxxxxxxxx

CHAT_ID=xxxxxxxxx

DATABASE_URL=postgres://....

REDIS_URL=xxxx

SCRAPE_INTERVAL=10

NODE_ENV=production

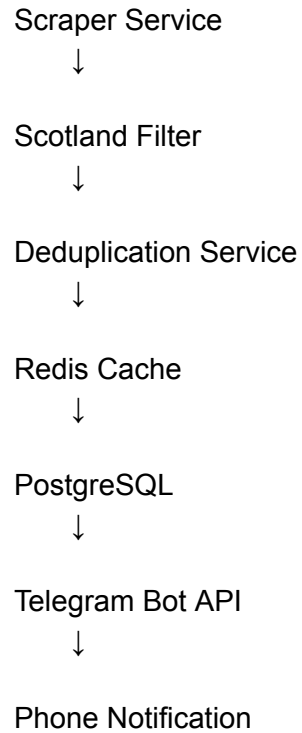
Final System Diagram

Amazon Jobs
Indeed
LinkedIn
Company Pages



Playwright/Axios





MVP Recommendation

For v1 specifically:

Node.js
TypeScript
Express
Playwright
PostgreSQL
Prisma
Telegram Bot API
Railway

This keeps your first version simple, cheap, and strong enough for portfolio use while allowing expansion later.